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Paper Code : OECCS701B Multimedia Systems

Time Allotted : 3 Hours

Full Marks :70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (I) A type of optical media?
- (II) Tag is used to create a numbered list in HTML?
- (III) In Audio & Video compression, the term RGB means
- (IV) State characteristic of a multimedia system?
- (V) Which image files are lossy format?
- (VI) DVD stands for
- (VII) The _____ defines the syntax of markup constructs and include additional definitions such as character entity references.
- (VIII) PDF/A format is
- (IX) What are multimedia software?
- (X) A smaller version of an image is called a
- (XI) _____ refers to information types synthesized by the computers.
- (XII) In Video Compression, an independent frame that is not related to any other frame is called

Group-B (Short Answer Type Question)

Answer any three of the following

[5 x 3 = 15]

2. Briefly explain JPEG image compression with block diagram. [5]
3. Define spatial and temporal redundancy? What is QoS? [5]
4. Write the components of multimedia. [5]
5. What is Text? What is rich text? What is glyph in Unicode standard? [5]
6. Write a short note about Video Conferencing. [5]

Group-C (Long Answer Type Question)

Answer any three of the following

[15 x 3 = 45]

7. (a) What is the purpose of synchronization? Describe the synchronization accuracy specification factors. [7]
 (b) Differentiate between CAV for hard disks and CLV for CDs. Describe Mode 1 and Mode 2 of CD-ROM. [5]
 (c) What do you mean by CCD? Why is it used? [3]
8. (a) What are lossy and lossless compression techniques? Give example of each kind. [4]
 (b) Briefly explain JPEG image compression with block diagram. [6]
 (c) Write the display sequence and transmission sequence for video frames. Why are they different? [5]
9. (a) What is sampling? Why is it required in multimedia? Define quantization error with a simple example. [5]
 (b) State the Nyquist sampling theorem. A communication channel can carry signal with frequency from 20 Hz to 20 KHz. Determine the sampling frequency. [5]
 (c) What are different image file formats? [5]
10. (a) Why synchronization is necessary in multimedia? Define spatial and temporal redundancy? [5]
 (b) What is CAV and CLV in case of spinning disk media? [5]
 (c) Describe briefly the display system technique and raster scan method. [5]
11. (a) What is Multimedia Database? Describe in brief how image database is constructed. [5]

- (b) What is K-d tree? Explain the structure of 2-dimension K-d tree by using following points: [7]
[(6, 1), (5, 5), (9, 6), (3, 6), (4, 9), (4, 0), (7, 9), (2, 9)]
- (c) Define QBIC. [3]

*** END OF PAPER ***